Patrick Selby

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EDUCATION

Grambling State University Expected May 2028

B.S. in Cybersecurity, Minor in Computer Information Systems | GPA: 3.88/4.00

Coursework: Software Security, Networking, Operating Systems, Computer Systems Architecture, Data Structures & Algorithms

WORK EXPERIENCE

CodePath | **Threat Detection Analyst** February 2026 – May 2026

- Investigated a Windows privilege-escalation scenario by correlating 4 event sources, reconstructing token theft and service installation, and mapping findings to MITRE ATT&CK T1134.001.
- Developed time-window correlation logic linking privilege adjustments and service installations to distinguish suspicious activity from routine administration.
- Documented attack evidence and correlation logic in a triage runbook to support repeatable investigations.

Hydroficient (via Extern) | **IoT Security Extern** February 2026 – March 2026

- Assessed connected-device security for an MQTT fleet using STRIDE threat modeling, prioritizing 2 high-severity risks by likelihood and impact.
- Implemented timestamps, monotonic counters, and HMAC checks to reject replayed messages, with per-device certificates over mutual TLS to authenticate connections.
- Developed an Isolation Forest telemetry detector to surface abnormal device traffic and prioritize investigation of suspicious behavior.

Grambling State University | **DOE CyberForce Blue Team Member** November 2025

- Defended enterprise and industrial-control infrastructure in DOE CyberForce with a six-member team, maintaining 97% availability across 6 critical services and placing in the top 10% nationally.
- Investigated 23 host and network events with Wireshark, logs, and command-line tools, identifying 8 high-confidence incidents and remediating 14 security misconfigurations.
- Automated 3 recurring checks with Python and Bash, cutting environment-validation time from approximately 20 minutes to under 5 minutes during live adversarial activity.

PROJECTS & RESEARCH

ScanSafe | Python, Flask, Kotlin GitHub

Grambling State University AIoT Lab, Security Research Assistant

- Built ScanSafe to assess phishing risk without sending URLs to external reputation services, implementing a 22-rule Python engine, Flask prototype, and Kotlin companion with engine evaluation under 5 ms.
- Established a 28-URL evaluation baseline with 85% commodity-phishing detection, 30% sophisticated-attack detection, and about 8% false positives to expose coverage gaps.

CloudCampus | Python, AWS S3, Lambda, IAM GitHub

- Built CloudCampus to automate student-document processing through S3-triggered Lambda text extraction, processing 50+ PDFs and reducing a 3-hour manual workflow to under 30 minutes.
- Scoped a least-privilege IAM execution role to the required bucket and actions, restricting document-processing access to the resources the pipeline needed.

LEADERSHIP & ACTIVITIES

Grambling Cybersecurity Alliance | **Technical Development Chair** January 2026 – Present

- Organized 18 technical preparation sessions and wrote 10 walkthroughs, preparing 15 first-time competitors for 4 CTFs and hackathons.

ColorStack Grambling | **Events Committee Member** August 2025 – Present

- Co-organized 4 campus career events reaching 200+ students and reviewed 40 peers' resumes to support technical career preparation.

SKILLS

Programming & Systems: Python, Bash, Kotlin, Linux, Windows Server, Git/GitHub
Security Engineering: MITRE ATT&CK, log correlation, incident triage, STRIDE, Isolation Forest, technical runbooks
Networking & Cryptography: Wireshark, MQTT, HMAC, TLS/mTLS, per-device certificates
Cloud & Applications: AWS S3, Lambda, IAM, least privilege, Flask, automated validation